A Novel Aerosol Retrieval Method Based on Joint Polarization and Intensity Data of Synchronization Monitoring Atmospheric Corrector (SMAC) Onboard High-Spatial-Resolution GFDM Satellite

Bangyu Ge, Xia Wang, Zhengqiang Li, Ping Zhou, Guoyuan Li, Weizhen Hou, Zhenwei Qiu, Chenchao Xiao, Shuhan Liu, Qingxing Yue, Haofei Wang, and Yisong Xie

***Abstract*— Synchronization monitoring atmospheric corrector (SMAC) sensor is equipped with polarization and intensity data, providing the possibility of retrieving more accurate aerosol parameters for main sensor's atmospheric correction. In this study, a novel aerosol retrieval algorithm based on intensity and polarization data was proposed. First, we analyze and construct the ratio relationship of different intensity and polarization channels of SMAC on different surface types and analyze the correlation between these ratios and normalized difference vegetation index (NDVI) and scattering angle (SCA). Then, for the first time, high-precision surface prior knowledge for SMAC aerosol retrieval is constructed, and then, intensity and polarization data were used to retrieve high-precision aerosol products simultaneously. These results are compared and validated with Moderate Resolution Imaging Spectroradiometer (MODIS) and ground-based observations aerosol products, and aerosol optical depth (AOD) product comparison between the SMAC and MODIS showed that they had similar spatial distribution, scattered dots of them with a Pearson correlation coefficient (R) of 0.84 and a root-mean-square error (RMSE) of 0.11. Meanwhile, their differences were also analyzed. The validation between the ground-based sites and the SAMC retrieval results also showed a good performance with R of 0.88 and RMSE of 0.10. These results revealed that the new algorithm is an effective method for retrieving reliable AOD products for the main sensor's atmosphere correction.**

***Index Terms*— Aerosol retrieval, intensity, polarization, surface prior knowledge, synchronization monitoring atmospheric corrector (SMAC).**

I. Introduction

SURFACE reflectance (SR) retrieved from satellite remote sensing is an essential input parameter for qualitative and quantitative studies of Earth's surface monitoring [1], [2], [3], [4], [5], [6]. As we all know, due to atmospheric scattering and absorption, the signal received by satellite sensor contains both atmosphere and surface information. Thus, atmospheric correction is the only way to obtain SR and aerosol is one of the key input parameters for atmospheric correction [7], [8]. Usually, for low-to-medium spatial resolution sensors with sufficient band information, aerosol can be retrieved using its own spectral information [9]. However, for the high-spatial-resolution sensors, it is difficult to perform high-precision aerosol retrieval using their limited bands. In this case, aerosol can be obtained from the ground-based observation data, other satellites' aerosol products, and synchronous atmospheric correction (SAC) instruments [10], [11], [12], [13]. Among them, aerosol from the SAC sensors has the best performance in terms of spatial–temporal matching.

Until now, various types of SAC sensors onboard different satellite platforms have been launched. In November 2000, the first SAC sensor linear etalon imaging spectral array/atmospheric corrector (LAC) onboard Earth Observing 1 (EO-1) was successfully launched, which is a hyperspectral imager and used to correct the images from the Advanced Land Imager [14]. Thermal and Near-Infrared Sensor for Carbon Observation Cloud and Aerosol Imager (TANSO-CAI) and enhanced TANSO-CAI-2 onboard Greenhouse Gases Observing Satellite (GOSAT) and GOSAT-2 were launched in January 2009 and October 2018, respectively, which are multispectral observation instruments designed to detect cloud and aerosol results for main payload atmospheric correction [15].

Received 14 September 2024; revised 24 January 2025; accepted 4 March 2025. Date of publication 11 March 2025; date of current version 24 March 2025. This work was supported in part by the National Outstanding Youth Foundation of China under Grant 41925019, in part by the National Natural Science Foundation of China under Grant 42105134, and in part by the National Civil Aerospace Pre-Research Project under Grant D030104. *(Corresponding authors: Zhengqiang Li; Bangyu Ge.)*

Bangyu Ge, Xia Wang, Ping Zhou, Guoyuan Li, Chenchao Xiao, Shuhan Liu, and Qingxing Yue are with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources of the People's Republic of China, Beijing 100048, China (e-mail: geby@lasac.cn; wangx@lasac.cn; zhoup@lasac.cn; ligy@lasac.cn; xiaocc@lasac.cn; liush@lasac.cn; yueqx@lasac.cn).

Zhengqiang Li, Weizhen Hou, and Yisong Xie are with the State Environment Protection Key Laboratory of Satellite Remote Sensing, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China (e-mail: lizq@radi.ac.cn; houwz@radi.ac.cn; xieys@aircas.ac.cn).

Zhenwei Qiu is with the Key Laboratory of Optical Calibration and Characterization, Hefei Institute of Physical Science, Chinese Academy of Sciences, Hefei 230031, China (e-mail: zwqiu@aiofm.ac.cn).

Haofei Wang is with the Innovation Center for Fengyun Meteorological Satellite (FYSIC), Key Laboratory of Radiometric Calibration and Validation for Environmental Satellites, National Satellite Meteorological Center (National Center for Space Weather), China Meteorological Administration, Beijing 100101, China (e-mail: wanghf@cma.gov.cn).

Digital Object Identifier 10.1109/TGRS.2025.3549452

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.

In August 2014, WorldView-3 was launched and equipped with an SAC sensor Clouds, Aerosols, water Vapor, Ice, and Snow (CAVIS), which has 12 bands from 405 to 2245 nm [16]. In July 2020, the first civil SAC sensor synchronization monitoring atmospheric corrector (SMAC) with polarization detection equipped on Chinese high-spatial-resolution satellite Gao Fen Duo Mo (GFDM) was successfully launched, SMAC has five polarized bands and three nonpolarized bands, which can be used to retrieve aerosol and water vapor for the main payload atmospheric correction [10]. In September 2020, the polarized scanning atmospheric corrector (PSAC) onboard Chinese HuanjingJianzai-2 (HJ-2) A/B satellites were launched, same as the SMAC, which is also with polarized channel and designed to retrieve the atmospheric correct parameters [17].

Aerosols retrieved from satellite remote sensing data have been developed for over 50 years; however, due to the complexity of surface–atmosphere coupling, high-quality retrieval of aerosols over land still faces challenges. Aerosol retrieval based on satellite remote sensing data is an ill-posed problem, and there are two main issues that need to be addressed: one is the prior estimation of the surface and the other is the prior estimation of the aerosol. Different aerosol retrieval algorithms have been developed based on intensity and polarization data. Dark target (DT) and deep blue (DB) are the two representative aerosol retrieval algorithms developed for intensity data and are the official algorithms of Moderate Resolution Imaging Spectroradiometer (MODIS). DT was proposed by Kaufman et al. [18], and the basic principle is that in DT (i.e., vegetated and dark-soiled) regions, the SR of visible channels can be estimated by the shortwave infrared (SWIR) band [19]. The DB algorithm was proposed by Hsu et al. [20], which utilizes the characteristic of the blue bands being less affected by the bright surfaces (i.e., arid and semiarid areas) to retrieve aerosols, and the prior SR databases of blue band were conducted by using the minimum reflectivity technique. MODIS DT and DB aerosol products have been validated to have good accuracy in global regions and are widely used in various fields [21]. Meanwhile, DT and DB algorithms were improved and applied to other sensors such as Visible Infrared Imaging Radiometer Suite (VIIIRS), Advanced Himawari Imager (AHI), and Medium Resolution Spectral Imager (MERSI) [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34]. However, there are some shortcomings of the DT and DB algorithm, and DT has the disadvantage of only being applicable to dark surfaces and requiring the SWIR band. DB constructing SR databases requires a large amount of historical data, especially for areas with high cloud coverage, and its aerosol optical depth (AOD) products have been validated with low accuracy in some regions [35]. To address these issues, Ge et al. [36] proposed a visible SR relationships (VISRRs) algorithm, which needs neither an SWIR band nor an SR database and can retrieve AOD over dark and bright land cover. VISRR has been applied to directional polarimetric camera (DPC) intensity data onboard Chinese Gaofen-5 series satellites, and the aerosol validation results in the study area show a good performance.

Compared to intensity data, the polarization signal is more sensitive to aerosol, many satellite polarimetric sensors, such as polarization and directionality of the Earth's reflectances (POLDER), DPC, PSAC, particulate observing scanning polarimeter (POSP), multiview multichannel multipolarization imaging (3MI), and MultiAngle Imager for Aerosols (MAIA), have been launched or to be launched for aerosol monitoring [37], [38]. Meanwhile, various aerosol retrieval algorithms based on polarization data have been developed. For example, based on the POLDER's polarization data, Deuzé et al. [39] used the bidirectional polarization distribution function (BPDF) model and assumed spectral neutrality of the red and infrared surface polarized reflectance (SPR) to retrieve the fine-mode aerosol. Wang et al. [40] constructed the red and infrared band SPR relationship changed by the normalized difference vegetation index (NDVI) and scattering angle (SCA) and combined the BPDF model to retrieve the AOD. However, the accuracy of BPDF model is relatively low on some surface types [41]. To overcome this problem, Ge et al. [42] abandoned using the BPDF model and proposed a spectral neutrality of SPR (SNOSPR) algorithm to retrieve the fine-mode aerosol. Another common polarized aerosol retrieval algorithm similar to DT, the visible and infrared SPRs, is estimated by the SWIR band [43]. However, compared to intensity SWIR data, polarization SWIR data are more susceptible to atmospheric influences.

Thus, in this study, a new aerosol retrieval algorithm for SMAC is proposed. The new algorithm analyzes and constructs the ratio relationship of different intensity and polarization channels of SMAC on different surface types and analyzes the correlation between these ratios and NDVI and SCA. Finally, for the first time, high-precision surface prior knowledge for SMAC aerosol retrieval is constructed, and then, intensity and polarization data were used to retrieve high-precision aerosol products simultaneously. These results are validated and compared with ground-based observations and MODIS aerosol products, and it is found that they have high accuracy and can be used for atmospheric correction of the main sensor. The study is structured as follows. In Section II, we introduce the datasets, while Section III presents the principle and methodology of the new algorithm. In Section IV, we compare and validate SMAC AOD products with ground-based observations and MODIS aerosol products. Section V presents the discussions, and Section VI summarizes the conclusion.

II. Data

A. SMAC Data

SMAC is designed to retrieve synchronous atmospheric parameters for atmospheric correction of the high spatial resolution main sensor. SMAC is capable of observing data at eight wavelengths of 490 nm (P), 550 and 670 nm (P), 870 nm (P), 910 and 1610 nm (P), and 2250 nm (P); the P in bracket represents the ability to observe polarization signals. The specific wavelength settings and their main applications are shown in Table I. SMAC operates simultaneously with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.

TABLE I
WAVELENGTH SETTING AND MAIN APPLICATIONS OF SMAC

Channel	Centre wavelength ±Spectral bandwidth (nm)	Polarization I/Q/U	Main application
1	490 ± 20	Yes	Aerosol retrieval/ clouds detection
2	550 ± 20	-	Snow detection
3	670 ± 20	Yes	Aerosol retrieval
4	870 ± 40	Yes	Aerosol/ Water vapor retrieval
5	910 ± 20	-	Water vapor retrieval
6	1380 ± 40	-	Cirrus clouds detection
7	1610 ± 60	Yes	Aerosol retrieval/ Snow detection
8	2250 ± 80	Yes	Aerosol retrieval

TABLE II
DATASETS OF MOD04 USED IN THIS STUDY

Product	Scientific Data Set (SDS) name (Contents)
MOD04 (C 6.1)	AOD_550_Dark_Target_Deep_Blue_Combined (Combined DT, DB AOD at 0.55 um)
	AOD_550_Dark_Target_Deep_Blue_Combined_QA_Flag (Combined DT, DB Aerosol Confidence Flag)
	Deep_Blue_Cloud_Fraction_Land (Cloud fraction from DB Aerosol cloud mask)

the main sensor using a dual-pixel observation mode, without forming images. It captures data at a dense rate of 10 frames/s, with a spatial resolution of approximately 6.7 km and a swath width of about 14 km [10], [12]. For this study, intensity and polarization data from SMAC, spanning nine months (from July 2020 to March 2021), were used for global aerosol retrieval.

B. MODIS Aerosol Product

MODIS Collection 6.1 aerosol products (MOD04/MYD04, where *O* is for Terra and *Y* is for Aqua) have been widely validated globally, have high accuracy, and are often used for comparison with other satellite products [44]. The DTB AOD products are composed exclusively of high-quality DT and DB data over both bright and dark surfaces. According to the study by Li et al. [45], the accuracy of the DTB products is generally better than the DT and DB AOD products. In this study, the datasets (as shown in Table II) of MOD04 aerosol products were used to compare with SMAC AOD. The spatial resolution of these datasets is 10 km.

C. Ground-Based Aerosol Product

Aerosols from ground-based observations [such as AERosol RObotic NETwork (AERONET) and Sun–Sky Radiometer Observation Network (SONET)] have high accuracy and are usually used as truth values for the validation of satellite remote sensing aerosol products, and the temporal resolution is about 15 min [46], [47]. In this study, AOD observations at 440, 500, and 670 nm from AERONET Version 3 Level 1.5 and SONET Level 1.5 were used to validate the AOD (550 nm) retrieved by SMAC.

D. Land Cover Type and Elevation Products

In this study, MODIS Land Cover Type (MCD12Q1) version 6 product was used to determine surface types; according to the International Geosphere Biosphere Program (IGBP) classification, global is divided into 17 surface types. The spatial resolution of MCD12Q1 is 500 m. In addition, Global Multiresolution Terrain Elevation Data 2010 (GMTED2010) with 1-km spatial resolution was used to determine the altitude.

III. PRINCIPLES AND METHODS

Assuming that the Earth's surface is Lambertian and homogeneous, the intensity and polarization signals received by satellite sensors in the visible-to-SWIR wavelength can be expressed as follows:

$$\rho_I^{\mathrm{TOA}} = \rho_I^{\mathrm{ATM}} + \frac{\mathrm{T}_I^{\downarrow} * \mathrm{T}_I^{\uparrow} * \mathrm{SR}}{1 - S * \mathrm{SR}} \tag{1}$$

$$\rho_P^{\mathrm{TOA}} = \rho_P^{\mathrm{ATM}} + \mathrm{T}_P^{\downarrow} * \mathrm{T}_P^{\uparrow} * \mathrm{SPR} \tag{2}$$

where ρ_I^{TOA} and ρ_P^{TOA} represent the intensity and polarization reflectance at the top of atmosphere (TOA), respectively; ρ^{ATM} and ρ_P^{ATM} represent intensity and polarization atmospheric path reflectance, respectively; and $\mathrm{T}_I^{\downarrow} * \mathrm{T}_I^{\uparrow}$ and $\mathrm{T}_P^{\downarrow} * \mathrm{T}_P^{\uparrow}$ represent downward and upward atmospheric transmittance, respectively. S represents atmospheric spherical albedo; SR and SPR represent surface reflectance and surface polarized reflectance, respectively; subscript I represents the intensity; and P represents the polarization. From the equation, it can be seen that ρ^{TOA} couples signals from the atmosphere and the surface. In order to obtain information about the atmosphere or the surface, it is necessary to perform atmosphere–surface decoupling. However, satellite remote sensing retrieval of aerosols is an ill-posed problem because the observed parameters are always less than the unknown parameters. Therefore, prior knowledge of aerosols and the surface needs to be prepared before retrieval.

A. New Prior Surface Estimation Strategy

No matter whether aerosol retrieval is based on intensity or polarization data, prior surface knowledge estimation is a key problem that needs to be solved. As introduced in Section I, for intensity data, there are mainly two categories of prior surface knowledge estimation. One is based on SR relationships or SR ratio between different bands. For example, the DT method estimates the SR of visible through short-wave infrared bands, and the VISRR algorithm uses the SR ratio between visible bands. The second major category is based on SR or SR ratio databases, such as the DB algorithm, and this method requires a large amount of historical data to construct the databases. For polarization data, there are also two main categories of prior SPR knowledge estimation. One is based on the BPDF model, and another is SPR relationships between different bands

$$\mathrm{SR} = \frac{\rho_I^{\mathrm{TOA}} - \rho_I^{\mathrm{ATM}}}{\left(\rho_I^{\mathrm{TOA}} - \rho_I^{\mathrm{ATM}}\right) * S + \mathrm{T}_I^{\downarrow} * \mathrm{T}_I^{\uparrow}} \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.

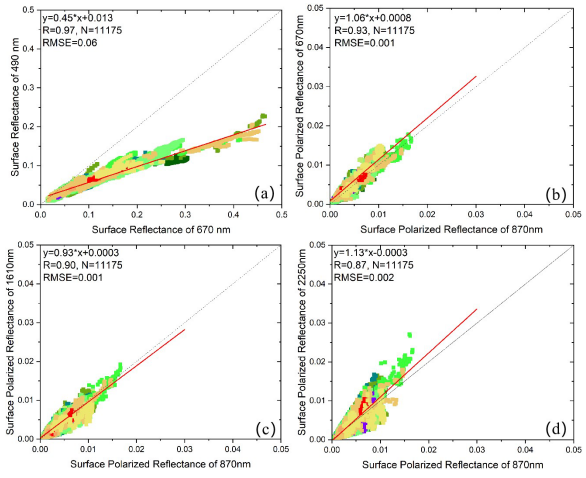


Fig. 1. (a) SR relationships between 490 and 670 nm. SPR relationships between (b) 670, (c) 1610, and (d) 2250 and 870 nm, the colors of the scattered dots represent different surface types.

$$\mathrm{SPR} = \frac{\rho_P^{\mathrm{TOA}} - \rho_P^{\mathrm{ATM}}}{\mathrm{T}_P^{\downarrow} * \mathrm{T}_P^{\uparrow}}. \tag{4}$$

Due to the lack of long-term and widely covered data of SMAC, in this study, a new combined prior SR and SPR ratio of SMAC was proposed. To obtain the SMAC prior SR and SPR, the MOD04 AOD products were used for atmosphere correction. The atmospheric correction process is as follows. First, spatially match between SMAC and MOD04 data according to latitude and longitude. Second, take the SMAC pixel as the center point and select the AOD values within the MODIS 5 × 5 window. Third, if all AOD values within the window exist and the average value is less than 0.1, then, assuming that the aerosol model is continental, the atmosphere parameters ρ_I^{ATM}, $\mathrm{T}_I^{\downarrow} * \mathrm{T}_I^{\uparrow}$, and S in (3) and ρ_P^{ATM} in (4) can be calculated by using the 6SV radiation transfer model, and $\mathrm{T}_P^{\downarrow} * \mathrm{T}_P^{\uparrow}$ can be calculated according to [48]. Finally, we can obtain the SR and SPR of SMAC.

Fig. 1 shows the SR and SPR relationships between different bands; Fig. 1(a) shows the SR relationships between 490 and 670 nm; and the slope and intercept of the fit curve are 0.45 and 0.013, respectively, with a Pearson correlation coefficient (R) of 0.97 and a root-mean-square error (RMSE) of 0.06. Fig. 1(b)–(d) shows the SPR relationships between 670, 1610, and 2250 and 670 nm, respectively. The slope of the fit curve is 1.06, 0.93, and 1.13 and the intercepts are 0.0008, 0.0003, and −0.0003, respectively, which are close to 1 and 0. R's are 0.93, 0.90, and 0.87 and RMSEs are 0.001, 0.001, and 0.002.

Although SR and SPR relationships have a high correlation across all surface types, there are differences, as shown in Fig. 2, and the colors of the scattered dots represent different surface types. Furthermore, we calculated the SR and SPR ratios of different surface types, as shown in Fig. 2. It can be found that the SR ratios of different surface types between 490 and 670 nm ($\mathrm{SR}_{490/670}$) are different and around 0.6, and the SPR ratios between 670, 1610, 2250, and 870 nm ($\mathrm{SPR}_{670/870}$, $\mathrm{SPR}_{1610/870}$, $\mathrm{SPR}_{2250/870}$) are different and around 1.

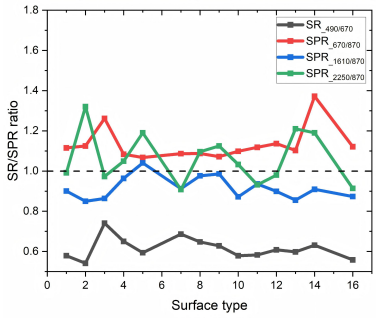


Fig. 2. SR ratio between 490 and 670 nm, and SPR ratios between 670, 1610, 2250, and 870 nm over different surface types.

TABLE III
PEARSON CORRELATION COEFFICIENTS BETWEEN SR/SPR RATIOS AND NDVI/SCA

NDVI/SCA / SR or SPR ratio	$\mathrm{NDVI}_{670/870}$	$\mathrm{NDVI}_{870/1610}$	$\mathrm{NDVI}_{870/2250}$	SCA
$\mathrm{SR}_{490/670}$	0.61	-0.62	-0.63	-0.03
$\mathrm{SPR}_{670/870}$	0.20	-0.26	-0.24	0.41
$\mathrm{SPR}_{1610/870}$	-0.09	0.13	0.13	-0.43
$\mathrm{SPR}_{2250/870}$	-0.31	0.34	0.32	-0.34

In addition, we analyzed the correlation between SR, SPR ratio versus NDVI(s), and SCAs. The traditional NDVI_670/870, calculated by red and near-infrared (NIR) TOA reflectance, is easily influenced by the atmosphere [49]. Thus, in this study, we introduced the NIR-SWIR vegetation index(s) ($\mathrm{NDVI}_{870/1610}$ and $\mathrm{NDVI}_{870/2250}$) for analysis, which is less affected by the atmosphere and has been widely used to indicate vegetation moisture condition [50]. The analysis results are shown in Table III. It can be found that NDVI(s) has a relatively high correlation with $\mathrm{SR}_{490/670}$, but a low correlation with $\mathrm{SPR}_{670/870}$, $\mathrm{SPR}_{1610/870}$, and $\mathrm{SPR}_{2250/870}$; SCA has a low correlation with $\mathrm{SR}_{490/670}$, but a relatively high correlation with $\mathrm{SPR}_{670/870}$, $\mathrm{SPR}_{1610/870}$, and $\mathrm{SPR}_{2250/870}$, which is consistent with Litvinov et al.'s [51] research.

Based on the above analysis, in this study, as shown in Tables IV–VII, $\mathrm{SR}_{490/670}$ is considered to be related to surface type, and $\mathrm{NDVI}_{870/2250}$, $\mathrm{SPR}_{670/870}$, $\mathrm{SPR}_{1610/870}$, and $\mathrm{SPR}_{2250/870}$ are considered to be related to surface type and SCA.

B. Aerosol Retrieval Based on Intensity and Polarization Data Simultaneously

In addition to prior surface estimation, aerosol models are another important prior knowledge for satellite aerosol remote sensing retrieval. In this study, six combinations with the highest appearance probability of China fundamental aerosol models, i.e., urban polluted and summer fly ash, secondary polluted and summer fly ash, combined polluted, and winter fly ash, continental background and background dust, continental background and winter fly ash, and continental background and transported dust [52], and five global aerosol modes, i.e., continental, moderately absorbing/developing, absorbing/smoke, nonabsorb/urban-ind, and background desert [53], were adapted for aerosol retrieval.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
VARIATION IN $SR_{_490/670}$ WITH SURFACE TYPE AND $NDVI_{_870/2250}$

Surface type / $NDVI_{_870/2250}$	1	2	3	4	5	7	8	9	10	11	12	13	14	16
[-1.0, -0.4)	0.77	0.54	0.71	0.75	0.77	0.80	0.72	0.64	0.74	0.60	0.80	0.64	0.63	0.56
[-0.4, 0.0)	0.64	0.54	0.74	0.66	0.65	0.69	0.68	0.66	0.61	0.60	0.62	0.64	0.63	0.56
[0.0 ,1.0]	0.50	0.45	0.74	0.57	0.54	0.59	0.57	0.58	0.55	0.50	0.51	0.60	0.63	0.56

When the prior knowledge of surface and aerosol model has been prepared, different from the traditional method, this usually retrieves the aerosol parameters using the single intensity or polarization data. In this study, a new method of aerosol retrieval based on SMAC intensity and polarization data simultaneously was proposed. To save the computation time, lookup tables (LUTs) were built by using the 6SV radiation transfer model; the input parameters of the 6SV RT model included the spectral response functions of SMAC 490, 670, 865, 1610, and 2250 nm; 11 aerosol models solar, and view zenith angle ranging from 0° to 80°, with an interval of 6°, relative azimuth angle ranging from 0° to 180° with an interval of 12°; and 11 AOD values at 550 nm (0, 0.01, 0.25, 0.5, 1, 1.5, 2, 2.5, 3, 4, and 5). Then, the atmospheric parameters vary with observational and atmospheric scenarios can be obtained and LUTs are constructed. When the LUTs and related auxiliary data were ready, the novel algorithm could be performed based on SMAC intensity and polarization data. All procedures were applied to the individual pixels, and the output result at each pixel was recorded. The schematic flowchart is shown in Fig. 3, and the specific retrieval steps are described as follows.

1) Convert SMAC L1 radiance data to TOA reflectance, including $\rho^{\mathrm{TOA}}_{I,490}$, $\rho^{\mathrm{TOA}}_{P,490}$, $\rho^{\mathrm{TOA}}_{I,550}$, $\rho^{\mathrm{TOA}}_{I,670}$, $\rho^{\mathrm{TOA}}_{P,670}$, $\rho^{\mathrm{TOA}}_{I,870}$, $\rho^{\mathrm{TOA}}_{P,870}$, $\rho^{\mathrm{TOA}}_{I,1380}$, $\rho^{\mathrm{TOA}}_{I,1610}$, $\rho^{\mathrm{TOA}}_{P,1610}$, $\rho^{\mathrm{TOA}}_{I,2250}$, and $\rho^{\mathrm{TOA}}_{P,2250}$.
2) Determine cloud and land pixel, and when a pixel meets the following conditions, it will be judged as a cloud-contaminated pixel:

$$\rho^{\mathrm{TOA}}_{I,490} > 0.4 \ \& \ \rho^{\mathrm{TOA}}_{I,1380} > 0.0025 \ \& \ \mathrm{NDDI} < 0 \ \& \ \mathrm{NDSI} > 0.13 \tag{5}$$

where normalized difference dust index (NDDI) is calculated by $\rho^{\mathrm{TOA}}_{I,490}$ and $\rho^{\mathrm{TOA}}_{I,2250}$ and normalized difference snow index (NDSI) is calculated by $\rho^{\mathrm{TOA}}_{I,550}$ and $\rho^{\mathrm{TOA}}_{I,2250}$ [10]. The land pixels are judged using MCD12Q1 surface-type data.
3) When the pixel is identified as the clear and land pixel, we applied the Rayleigh scattering correction method proposed by Fraser et al. [54]. This method adjusts the LUT(s) by simulating varying Rayleigh optical depths based on changes in wavelength. The relationship between wavelength and surface target height is expressed as

$$\lambda(h = H) = \lambda(h = 0) * \exp\left(\frac{H}{34}\right) \tag{6}$$

where h represents the surface target height in kilometers and $h = 0$ is the height at sea level. Once the surface target height H is known, the corresponding wavelength can be adjusted accordingly. With the adjusted wavelength, new LUT(s) values can be interpolated from the LUT(s) computed at sea level [19]. For instance, if $H = 0.7$ km, $\lambda(h = 0) = 0.49$ μm; according to (6), $\lambda(h = 0.7) = 0.50$ μm. In this case, the LUT(s) values corresponding to a 0.5-μm wavelength can be obtained by interpolating between the entries at 0.49 and 0.67 μm.
4) Then, based on observation geometry angles and new LUT(s), atmospheric parameters can be interpolated for different aerosol models and AOD(s) [19]. Then, according to (1) and (2), $\mathrm{SR}^{\mathrm{cal}}_{490}(M,\tau)$, $\mathrm{SPR}^{\mathrm{cal}}_{490}(M,\tau)$, $\mathrm{SR}^{\mathrm{cal}}_{670}(M,\tau)$, $\mathrm{SPR}^{\mathrm{cal}}_{670}(M,\tau)$, $\mathrm{SR}^{\mathrm{cal}}_{870}(M,\tau)$, $\mathrm{SPR}^{\mathrm{cal}}_{870}(M,\tau)$, $\mathrm{SPR}^{\mathrm{cal}}_{1610}(M,\tau)$, $\mathrm{SR}^{\mathrm{cal}}_{2250}(M,\tau)$, and $\mathrm{SPR}^{\mathrm{cal}}_{2250}(M,\tau)$ can be obtained, where M represents the aerosol model and τ represents the AOD.
5) Calculate $\mathrm{NDVI}_{_870/2250}$, and according to the surface type, we can obtain $\mathrm{SR}_{_490/670}$ from Table IV. Then, we can construct cost function 1 as follows:

$$\mathrm{CF}_1 = \left(\frac{\mathrm{SR}^{\mathrm{cal}}_{490}(M,\tau)}{\mathrm{SR}^{\mathrm{cal}}_{670}(M,\tau)} - \mathrm{SR}_{_490/670}\right). \tag{7}$$

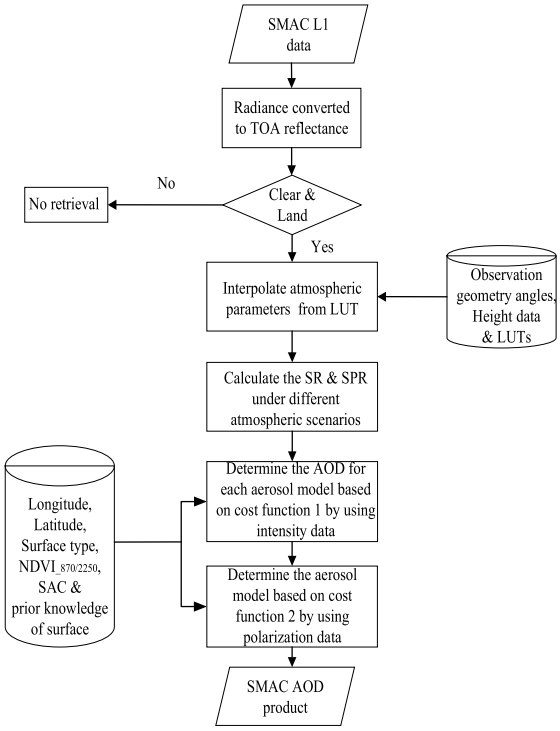


Fig. 3. Flowchart of the main steps of the new SMAC aerosol retrieval algorithm.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.

TABLE V
VARIATION IN $SPR_{_670/870}$ WITH SURFACE TYPE AND SCA

Surface type / SCA	1	2	3	4	5	7	8	9	10	11	12	13	14	16
[60°, 120°)	1.06	0.91	1.07	0.99	0.98	1.12	0.96	0.99	0.95	1.08	1.00	1.09	1.37	0.99
[120°, 140°)	1.10	1.16	1.25	1.19	1.21	1.07	1.04	1.04	1.05	1.09	1.04	1.09	1.37	0.99
[140°,180°]	1.18	1.44	1.64	1.19	1.24	1.22	1.32	1.21	1.22	1.83	1.24	1.31	1.37	1.29

TABLE VI
VARIATION IN $SPR_{_1610/870}$ WITH SURFACE TYPE AND SCA

Surface type / SCA	1	2	3	4	5	7	8	9	10	11	12	13	14	16
[60°, 120°)	1.00	0.91	1.22	1.12	1.15	0.83	1.14	1.02	1.05	0.98	0.99	0.87	0.91	1.00
[120°, 140°)	0.87	0.84	0.81	0.78	0.86	0.93	1.00	1.03	0.90	0.94	1.01	0.87	0.91	1.00
[140°,180°]	0.83	0.75	0.53	0.80	0.85	0.79	0.77	0.82	0.76	0.83	0.81	0.67	0.91	0.71

TABLE VII
VARIATION IN $SPR_{_2250/870}$ WITH SURFACE TYPE AND SCA

Surface type / SCA	1	2	3	4	5	7	8	9	10	11	12	13	14	16
[60°, 120°)	1.17	1.70	1.73	1.23	1.36	1.12	1.27	1.21	1.22	1.20	1.10	1.22	1.19	1.01
[120°, 140°)	0.95	1.27	0.80	0.94	0.95	0.91	1.08	1.12	0.96	0.90	1.17	1.22	1.19	1.01
[140°,180°]	0.87	0.50	0.53	0.82	0.85	0.89	0.96	1.06	1.01	0.99	0.84	1.02	1.19	0.79

We can make $CF_1 = 0$ by interpolating τ, then different τ's for different aerosol models can be obtained.

Due to the polarization signal being more sensitive to aerosol, in this study, the polarization data were used to determine the best aerosol model. According to the SCA, we can obtain $SPR_{_670/870}$, $SPR_{_1610/870}$, and $SPR_{_2250/870}$ in Tables V–VII, respectively. Thus, we can construct cost function 2 as follows:

$$CF_2 = \sqrt{\sum_{i=1}^{N}\left(\frac{SPR_{\lambda_i}^{cal}(M)}{SPR_{870}^{cal}(M)} - SPR_{_\lambda_i/870}\right)^2} \quad (8)$$

where $N = 3$ and λ_1–λ_3 represent 670, 1610, and 2250 nm, respectively. When the aerosol model makes CF_2 minimum, the aerosol model and corresponding τ are the final results.

IV. RESULT VALIDATION

A. Compared With MODIS Product

MOD04 merged DT and DB AOD products (DTB) were used to compare with the SMAC AOD, and the spatial resolution of DTB and SMAC AOD is 10 and 6.7 km, respectively. First, use HDF-EOS to GeoTIFF Conversion Tool (HEG) to perform geometric correction on MOD04 and MOD02HKM products. Second, match the SMAC and MODIS AOD products based on latitude and longitude. Figs. 4–6 show the comparison of SMAC and MODIS AOD under heavy, moderate, and slightly pollution conditions, respectively. In these three cases, the time difference between the SMAC and MODIS observations is within 30 min.

Fig. 4(a) shows a heavy aerosol pollution case, and there is a pollution belt from northwest to southeast, with AOD values mostly above 1.25. The circle inside the red box represents

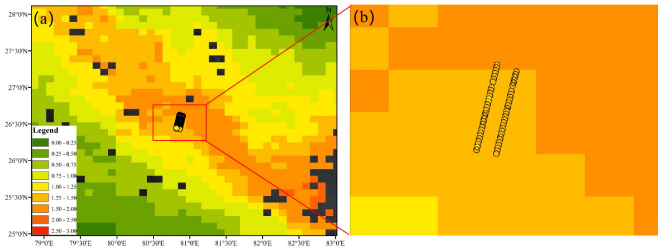


Fig. 4. (a) AOD spatial distribution of MODIS DTB and SMAC (circle) in 2020-12-04, the base image is MODIS true color image, and the area (red box) is suffering from heavy aerosol pollution. (b) Enlarged image of it.

SMAC AOD, as shown in the enlarged image of Fig. 4(b), and most of the SMAC AOD is around 1.33 and MODIS DTB AOD is around 1.37; the mean relative error is about 3%. Fig. 5(a) shows a moderate aerosol pollution case, most of the AOD values in the northern part of the region are less than 0.75, most of the SMAC AOD is around 0.71, and MODIS DTB AOD is around 0.75; the mean relative error is about 5%. Fig. 6(a) shows a slightly aerosol pollution case, most of the AOD values in the northeast part of the region are less than 0.5, most of the SMAC AOD is around 0.41, and MODIS DTB AOD is around 0.5; the mean relative error is about 18%. These comparison results show that when the overpass times are close, the two AOD products exhibit good consistency in spatial distribution.

Although the accuracy of the MODIS and SMAC AOD products is relatively similar in some cases, as shown in Figs. 4–6, this does not mean that the MODIS AOD product can replace the SMAC product. First, the spatial resolution of the 10-km MODIS AOD product is coarser than that of the 6.7-km SMAC AOD product, leading to weaker spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.

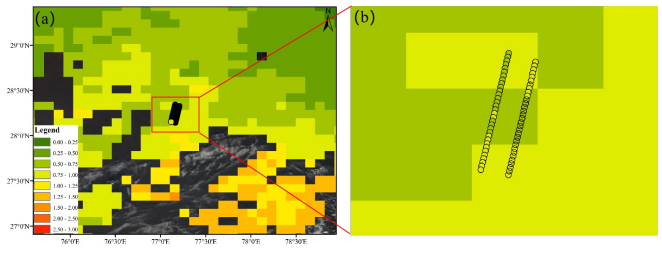


Fig. 5. (a) AOD spatial distribution of MODIS DTB and SMAC (circle) in 2020-11-18, the base image is MODIS true color image, and the area (red box) is suffering from moderate aerosol pollution. (b) Enlarged image of it.

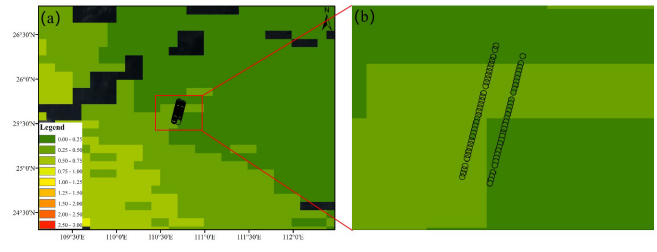


Fig. 6. (a) AOD spatial distribution of MODIS DTB and SMAC (circle) in 2021-02-20, the base image is MODIS true color image, and the area (red box) is suffering from slightly aerosol pollution. (b) Enlarged image of it.

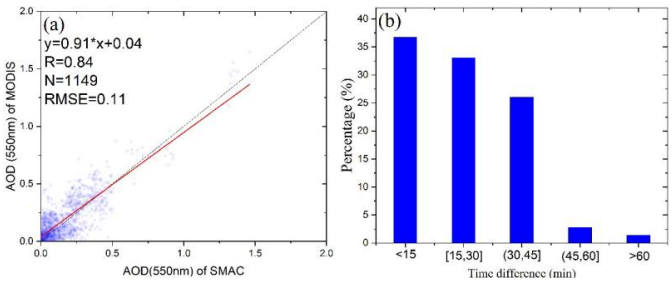


Fig. 7. (a) Comparison of AOD (550 nm) products between SMAC and MODIS DTB. (b) Percentage difference in time between SMAC and MODIS.

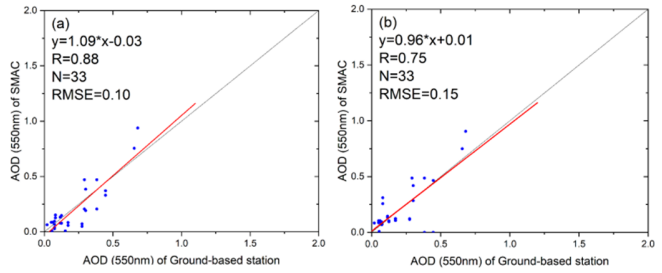


Fig. 8. Validation of AOD (550 nm) products between SMAC and ground-based observations. (a) New prior surface estimation strategy proposed in this study. (b) Mean SR/SPR ratio method.

detail information of aerosols. Second, there are differences in overpass times between the two datasets. Fig. 7(a) illustrates the comparison between MODIS and SMAC AOD products. The comparison method is as follows.

First, time matching will be performed, and then, all SMAC AOD within a radius of 5 km centered on MODIS pixels will be averaged as comparison results. At the same time, although multiple threshold values were applied for SMAC cloud detection, it does not produce an image, preventing the use of texture information for precise cloud identification, particularly for thin or fragmented clouds [10]. In order to shield the influence of the clouds, the cloud fraction from DB aerosol cloud mask were used in this study; when the mean fraction of all pixels within the MODIS 5 × 5 window is greater than 0.1, the pixels will be discarded. In future atmospheric correction research for the GFDM main sensor, spectral and texture information from the main sensor can be utilized for cloud detection, enabling the selection of more accurate SMAC AOD values. Fig. 7(a) shows the comparison results, the slope and intercept of fit line are 0.91 and 0.04, respectively, and the corresponding R and RMSE values are 0.84 and 0.11, respectively. Overall, the comparison results suggest that SMAC AOD is in good agreement versus MODIS products.

However, some discrepancies remain, such as underestimation at low AOD values and overestimation at high AOD values. These discrepancies may be attributed to the time differences between the two datasets. Typically, to capture aerosol variability, ground-based observations are taken at intervals of 15 min [46], however, as shown in Fig. 7(b), for the comparison between SMAC and MODIS, only 36.73% of cases have a time difference within 15 min, 33.07% fall between 15 and 30 min, 26.02% are between 30 and 45 min, and 4.18% exceed 45 min.

B. Validation With Ground-Based Data

The ground-based AERONET and SONET sites were used to validate SMAC AOD products. Due to the lack of ground-based measurements with 550-nm AOD products, a second-order polynomial fitting method was chosen to calculate the 550-nm AOD using the 440-, 500-, and 675-nm values [55]. In terms of time matching, the average value of ground-based observations within 30 min before and after the satellite overpass was chosen. In terms of spatial matching, all the valid SMAC retrieval results are averaged within 0.5° of latitude and longitude (about 50 km) centered on the ground site for validation. Meanwhile, as described in Section IV-A, the cloud fraction from DB aerosol cloud mask was used to shield the influence of the clouds. Fig. 8(a) shows the validation results between SMAC AOD products and ground-based observations. Due to the SAMC not imaging and being affected by clouds, there are 33 match-up points, and the slope and intercept of fit line are 1.09 and 0.03, respectively; the corresponding R and RMSE values are 0.88 and 0.10, respectively. Overall, the validation results suggest that SMAC AOD using the novel SMAC aerosol retrieval algorithm proposed in this study has a good performance and can be used for the main senor atmosphere correction.

C. Comparison of Different Prior Surface Estimation Strategies

Errors in prior surface estimation can cause AOD retrieval errors to increase by as much as tenfold, highlighting the critical importance of accurate prior surface knowledge [56]. To evaluate the effectiveness of the prior surface estimation strategy proposed in this study, we compare it with a fixed SR/SPR ratio approach, with values calculated from Tables IV to VII. Fig. 8(b) presents the validation results for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.

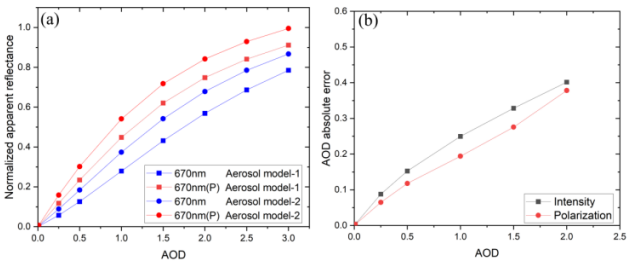


Fig. 9. (a) Simulated 0.67-μm normalized apparent reflectance for polarization and intensity as a function of AOD under different aerosol models. (b) Absolute errors when assuming Aerosol Model-1 as the true model and selecting Aerosol Model-2 using intensity and polarization data.

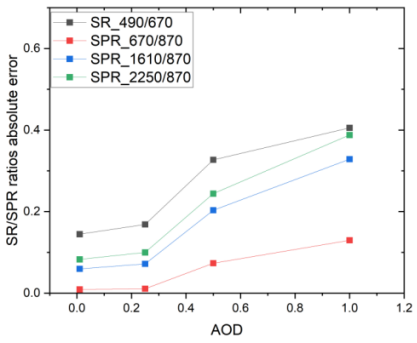


Fig. 10. Simulated absolute errors of SR and SPR ratios under different AOD conditions.

the fixed SR/SPR ratio method. Compared to the results from this study [Fig. 8(a)], R decreases from 0.88 to 0.75, and the RMSE increases from 0.1 to 0.15, indicating a decline in overall accuracy. This decline is due to the inability of the fixed SR/SPR ratio method to accurately estimate SR. For example, at the AERONET USDA_ALARC site, the ground-based AOD value is 0.08, while the AOD retrieved using the method proposed in this study is 0.12. In contrast, the fixed SR/SPR ratio method yields an AOD retrieval of 0.26, with the absolute error increasing from 0.04 to 0.18. These results clearly demonstrate that using a fixed SR/SPR ratio for all surface types introduces significant errors in some cases, whereas the method proposed in this study achieves superior accuracy.

V. Discussion

A. Sensitivity Analysis of Aerosol Models to Polarization and Intensity

Polarization data play a crucial role in aerosol retrievals, as it significantly improves the accuracy of aerosol property estimations, particularly in distinguishing aerosols from SR [38]. In the method proposed in this study, polarization information is utilized to select the optimal aerosol model. To test the sensitivity of polarization and intensity to aerosol models, theoretical experiments were conducted. The simulation conditions included a solar zenith angle of 42°, an observation zenith angle of 24°, a relative azimuth angle of 120°, and a vegetation surface with SR of 0.0328 and SPR of 0.0045 [57]. Two aerosol models were considered: Aerosol_mode-1 (urban polluted and summer fly ash) and Aerosol_mode-2 (combined polluted and winter fly ash) [52]. Fig. 9(a) shows the simulated variations of 0.67-μm normalized apparent reflectance for intensity and polarization under different aerosol types as AOD changes. It can be observed that the sensitivity of polarized apparent reflectance to differences in aerosol models is significantly higher than that of intensity. Fig. 9(b) further illustrates that when Aerosol_mode-1 is the true model, the absolute errors from incorrectly selecting Aerosol_mode-2 are much lower when polarization signals are used compared to intensity.

B. Analysis of the Impact of DTB AOD Accuracy on Prior SR and SPR Ratios

In this study, the MOD04 DTB AOD products were utilized for atmospheric correction to construct the SMAC prior SR and SPR ratios. As shown in Section III-A, only MODIS DTB mean AOD values below 0.1 within a 5 × 5 window were used for the atmospheric correction of SMAC. To assess the influence of MODIS DTB AOD accuracy on SR/SPR ratios, the following analysis was conducted. Using the same observation angles and surface settings as in Section V-A, we simulated SMAC apparent reflectance under varying AOD conditions (0.1) with a continental aerosol model. The simulated apparent reflectance was then atmospheric corrected using an assumed AOD value of 0.1 to derive the SR^{cal} and SPR^{cal}. The SR and SPR ratios were then calculated and compared to the true values, with their absolute differences evaluated. As shown in Fig. 10, the absolute errors of SR/SPR ratios are relatively small near AOD = 0.1, generally staying within 0.2. Among the ratios, $SPR_{670/870}$ shows the smallest error, while $SR_{490/670}$ exhibits the largest. However, as AOD increases, the absolute errors of SR/SPR ratios also grow larger, indicating that deviations in MODIS DTB AOD accuracy can lead to increased uncertainty in SR/SPR ratios. The mean values of the constructed $SR_{490/670}$ and $SPR_{670/870}$, and $SPR_{1610/87}$ and $SPR_{2250/870}$ ratios in this study are approximately 0.6 and 1, respectively, which are consistent with previous atmospheric correction results based on AERONET sites [19], [40]. Additionally, improvements were made by incorporating variations with surface types, SCA, and NDVI, further enhancing their accuracy and reliability.

VI. Conclusion

SAMC is a synchronization atmospheric correction instrument, which is designed to retrieve atmosphere parameters for the main sensor's atmosphere correction. In this study, an innovative algorithm using the SMAC intensity and polarization data simultaneously to retrieve AOD over both dark and bright land cover was proposed. First, the new algorithm analyzes and constructs the ratio relationship of different intensity and polarization channels of SMAC on different surface types and analyzes the correlation between these ratios and NDVI and SCA. Then, for the first time, high-precision surface prior knowledge for SMAC aerosol retrieval is constructed, and then, intensity and polarization data were used to retrieve high-precision aerosol products simultaneously, i.e., the AOD of each aerosol model is determined using intensity data, and then, the optimal aerosol model is determined using polarization data. These results are validated and compared with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.

ground-based observations and MODIS aerosol products, and the outcomes are analyzed and discussed. It is found that they have high accuracy and can be used for atmospheric correction of the main sensor. When applying SMAC atmospheric parameters to correct the GFDM main sensor data, spatiotemporal matching is required based on the acquisition time and location to select the corresponding aerosol parameters. In addition to the SMAC, the new algorithm can be applied to similar sensors with intensity and polarized data, such as the advanced DPC/GF-5 (02), CM-1, DQ-1, DQ-2, and 3MI [37].

ACKNOWLEDGMENT

The authors would like to express their gratitude to the AERONET PI(s) and teams. They are grateful to China Centre for Resources Satellite Data and Application for providing SMAC data, NASA for providing the MODIS products data, and USGS for providing GMTED2010. They deeply appreciate the detailed comments and valuable suggestions from anonymous reviewers.

REFERENCES

[1] P. J. Sellers et al., "Remote sensing of the land surface for studies of global change: Models—Algorithms—Experiments," *Remote Sens. Environ.*, vol. 51, no. 1, pp. 3–26, Jan. 1995.

[2] C. O. Justice et al., "The Moderate Resolution Imaging Spectroradiometer (MODIS): Land remote sensing for global change research," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 4, pp. 1228–1249, Jul. 1998.

[3] K. Jia et al., "Global land surface fractional vegetation cover estimation using general regression neural networks from MODIS surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 4787–4796, Sep. 2015.

[4] L. Xu et al., "Time series analysis for global land cover change monitoring: A comparison across sensors," *Remote Sens. Environ.*, vol. 271, Mar. 2022, Art. no. 112905.

[5] W. Hou, J. Wang, X. Xu, J. S. Reid, and D. Han, "An algorithm for hyperspectral remote sensing of aerosols: 1. Development of theoretical framework," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 178, pp. 400–415, Jul. 2016.

[6] W. Hou, J. Wang, X. Xu, and J. S. Reid, "An algorithm for hyperspectral remote sensing of aerosols: 2. Information content analysis for aerosol parameters and principal components of surface spectra," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 192, pp. 14–29, May 2017.

[7] Y. J. Kaufman and C. Sendra, "Algorithm for automatic atmospheric corrections to visible and near-IR satellite imagery," *Int. J. Remote Sens.*, vol. 9, no. 8, pp. 1357–1381, Aug. 1988.

[8] E. F. Vermote and S. Kotchenova, "Atmospheric correction for the monitoring of land surfaces," *J. Geophys. Res., Atmos.*, vol. 113, no. D23, p. 23, Dec. 2008.

[9] E. Vermote, C. Justice, M. Claverie, and B. Franch, "Preliminary analysis of the performance of the Landsat 8/OLI land surface reflectance product," *Remote Sens. Environ.*, vol. 185, pp. 46–56, Nov. 2016.

[10] Z. Li et al., "Preliminary on-orbit performance test of the first polarimetric synchronization monitoring atmospheric corrector (SMAC) on-board high-spatial resolution satellite gao fen duo mo (GFDM)," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022. [Online]. Available: https://ieeexplore.ieee.org/document/9538398

[11] D. Bru et al., "Atmospheric correction of multi-spectral littoral images using a PHOTONS/AERONET-based regional aerosol model," *Remote Sens.*, vol. 9, no. 8, p. 814, Aug. 2017.

[12] L. Xu et al., "Synchronous atmospheric correction of high spatial resolution images from Gao Fen Duo Mo satellite," *Remote Sens.*, vol. 14, no. 17, p. 4427, Sep. 2022.

[13] S. Sartika et al., "Determining the precision of spectral patterns arising from atmospheric correction utilizing MODTRAN-FLAASH and 6S approaches on high-resolution SPOT-6 imagery," in *Proc. IEEE Int. Conf. Aerosp. Electron. Remote Sens. Technol. (ICARES)*, Oct. 2023, pp. 1–7.

[14] D. C. Reuter et al., "The LEISA/atmospheric corrector (LAC) on EO-1," in *Proc. Scanning Present Resolving Future IEEE Int. Geosci. Remote Sens. Symp.*, vol. 1, Jul. 2001, pp. 46–48.

[15] R. Imasu et al., "Greenhouse gases observing SATellite 2 (GOSAT-2): Mission overview," *Prog. Earth Planet. Sci.*, vol. 10, no. 1, p. 33, Jul. 2023.

[16] L. Barazzetti, F. Roncoroni, R. Brumana, and M. Previtali, "Georeferencing accuracy analysis of a single worldview-3 image collected over Milan," in *Proc. 23th ISPRS Congr.*, Jun. 2016, pp. 429–434.

[17] Z. Li et al., "In-orbit test of the polarized scanning atmospheric corrector (PSAC) onboard Chinese environmental protection and disaster monitoring satellite constellation HJ-2 A/B," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022. [Online]. Available: https://ieeexplore.ieee.org/document/9779761

[18] Y. J. Kaufman, D. Tanre, L. A. Remer, E. F. Vermote, A. Chu, and B. N. Holben, "Operational remote sensing of tropospheric aerosol over the land from EOS-MODIS," *J. Geophys. Res., Atmos.*, vol. 102067, no. D14, p. 17, Jul. 1997.

[19] R. C. Levy, L. A. Remer, S. Mattoo, E. F. Vermote, and Y. J. Kaufman, "Second-generation operational algorithm: Retrieval of aerosol properties over land from inversion of Moderate Resolution Imaging Spectroradiometer spectral reflectance," *J. Geophys. Res., Atmos.*, vol. 112, no. D13, p. 13211, Jul. 2007.

[20] N. C. Hsu, S.-C. Tsay, M. D. King, and J. R. Herman, "Aerosol properties over bright-reflecting source regions," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 557–569, Mar. 2004.

[21] Q.-X. Chen et al., "Evaluation of MODIS, MISR, and VIIRS daily level-3 aerosol optical depth products over land," *Atmos. Res.*, vol. 265, Jan. 2022, Art. no. 105810.

[22] B. Ge et al., "A dark target method for Himawari-8/AHI aerosol retrieval: Application and validation," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 381–394, Jan. 2019, doi: 10.1109/TGRS.2018.2854743.

[23] L. Gao, L. Chen, J. Li, C. Li, and L. Zhu, "An improved dark target method for aerosol optical depth retrieval over China from Himawari-8," *Atmos. Res.*, vol. 250, Mar. 2021, Art. no. 105399.

[24] V. Sawyer, R. C. Levy, S. Mattoo, G. Cureton, Y. Shi, and L. A. Remer, "Continuing the MODIS dark target aerosol time series with VIIRS," *Remote Sens.*, vol. 12, no. 2, p. 308, Jan. 2020.

[25] S. Jin et al., "Adapting the dark target algorithm to advanced MERSI sensor on the FengYun-3-D satellite: Retrieval and validation of aerosol optical depth over land," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8781–8797, Oct. 2021.

[26] X. Su et al., "Retrieval of total and fine mode aerosol optical depth by an improved MODIS dark target algorithm," *Environ. Int.*, vol. 166, Aug. 2022, Art. no. 107343.

[27] Y. Wu, M. de Graaf, and M. Menenti, "Improved MODIS dark target aerosol optical depth algorithm over land: Angular effect correction," *Atmos. Meas. Techn.*, vol. 9, no. 11, pp. 5575–5589, Nov. 2016.

[28] L. Yang, Y. Xue, J. Guang, H. Kazemian, J. Zhang, and C. Li, "Improved aerosol optical depth and Ångstrom exponent retrieval over land from MODIS based on the non-lambertian forward model," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 9, pp. 1629–1633, Sep. 2014.

[29] J. M. Jackson et al., "Suomi-NPP VIIRS aerosol algorithms and data products," *J. Geophys. Research: Atmos.*, vol. 118, no. 22, p. 12, Oct. 2013.

[30] N. C. Hsu, J. Lee, A. M. Sayer, W. Kim, C. Bettenhausen, and S. Tsay, "VIIRS deep blue aerosol products over land: Extending the EOS long-term aerosol data records," *J. Geophys. Res., Atmos.*, vol. 124, no. 7, pp. 4026–4053, Apr. 2019.

[31] J. Wei, Z. Li, Y. Peng, L. Sun, and X. Yan, "A regionally robust high-spatial-resolution aerosol retrieval algorithm for MODIS images over eastern China," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4748–4757, Jul. 2019.

[32] L. Yang et al., "Preliminary test of quantitative capability in aerosol retrieval over land from MERSI-II onboard FY-3D," *Nat. Remote Sens. Bull.*, vol. 26, no. 5, pp. 923–940, 2022, doi: 10.11834/jrs.20210286.

[33] L. Yang et al., "Global evaluation of Fengyun-3 MERSI dark target aerosol retrievals over land," *Int. J. Digit. Earth*, vol. 17, no. 1, pp. 1–24, Dec. 2024.

[34] X. Pei et al., "Applying the dark target aerosol algorithm to MERSI-II: Retrieval and validation of aerosol optical depth over the ocean," *Adv. Atmos. Sci.*, vol. 41, no. 12, pp. 2446–2463, Dec. 2024, doi: 10.1007/s00376-024-4032-y.

[35] L. Yang et al., "Evaluation and comparison of MODIS C6 and C6.1 deep blue aerosol products in arid and semi-arid areas of Northwestern China," *Remote Sens.*, vol. 14, no. 8, p. 1935, Apr. 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/8/1935

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.

[36] B. Ge et al., "An improved aerosol optical depth retrieval algorithm for multiangle directional polarimetric camera (DPC)," *Remote Sens.*, vol. 14, no. 16, p. 4045, Aug. 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/16/4045

[37] O. Dubovik et al., "Polarimetric remote sensing of atmospheric aerosols: Instruments, methodologies, results, and perspectives," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 224, pp. 474–511, Feb. 2018.

[38] W. Hou, Z. Li, J. Wang, X. Xu, P. Goloub, and L. Qie, "Improving remote sensing of aerosol microphysical properties by near-infrared polarimetric measurements over vegetated land: Information content analysis," *J. Geophys. Res., Atmos.*, vol. 123, no. 4, pp. 2215–2243, Feb. 2018.

[39] J. L. Deuzé et al., "Remote sensing of aerosols over land surfaces from POLDER-ADEOS-1 polarized measurements," *J. Geophys. Research: Atmos.*, vol. 106, no. D5, pp. 4913–4926, Mar. 2001.

[40] H. Wang, L. Yang, M. Zhao, W. Du, P. Liu, and X. Sun, "The normalized difference vegetation index and angular variation of surface spectral polarized reflectance relationships: Improvements on aerosol remote sensing over land," *Earth Space Sci.*, vol. 6, no. 6, pp. 982–989, Jun. 2019.

[41] B. Yang, H. Zhao, and W. Chen, "Semi-empirical models for polarized reflectance of land surfaces: Intercomparison using space-borne POLDER measurements," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 202, pp. 13–20, Nov. 2017.

[42] B. Ge et al., "An improved algorithm for retrieving high resolution fine-mode aerosol based on polarized satellite data: Application and validation for POLDER-3," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111894, doi: 10.1016/j.rse.2020.111894.

[43] F. Waquet et al., "Aerosol retrieval over land using a multiband polarimeter and comparison with path radiance method," *J. Geophys. Res., Atmos.*, vol. 112, no. D11, p. 11214, Jun. 2007.

[44] J. Wei, Z. Li, Y. Peng, and L. Sun, "MODIS collection 6.1 aerosol optical depth products over land and ocean: Validation and comparison," *Atmos. Environ.*, vol. 201, pp. 428–440, Mar. 2019.

[45] Y. Li, G. Shi, and Z. Sun, "Evaluation and improvement of MODIS aerosol optical depth products over China," *Atmos. Environ.*, vol. 223, Feb. 2020, Art. no. 117251.

[46] D. M. Giles et al., "Advancements in the aerosol robotic network (AERONET) version 3 database—Automated near-real-time quality control algorithm with improved cloud screening for sun photometer aerosol optical depth (AOD) measurements," *Atmos. Meas. Techn.*, vol. 12, no. 1, pp. 169–209, Jan. 2019, doi: 10.5194/amt-12-169-2019.

[47] Z. Q. Li et al., "Comprehensive study of optical, physical, chemical, and radiative properties of total columnar atmospheric aerosols over China: An overview of Sun–Sky radiometer observation network (SONET) measurements," *Bull. Amer. Meteorolog. Soc.*, vol. 99, no. 4, pp. 739–755, Apr. 2018.

[48] F. Waquet, J.-F. Léon, B. Cairns, P. Goloub, J.-L. Deuzé, and F. Auriol, "Analysis of the spectral and angular response of the vegetated surface polarization for the purpose of aerosol remote sensing over land," *Appl. Opt.*, vol. 48, no. 6, p. 1228, 2009.

[49] B. Ge et al., "An impact study of NDVI on the BPDF model under different atmosphere and multi-angles conditions," in *Proc. AOPC Opt. Sens. Imag. Technol.*, Nanjing, China, Dec. 2019, doi: 10.1117/12.2547899.

[50] L. Ji, L. Zhang, B. K. Wylie, and J. Rover, "On the terminology of the spectral vegetation index (NIR-SWIR)/(NIR + SWIR)," *Int. J. Remote Sens.*, vol. 32, no. 21, pp. 6901–6909, Nov. 2011.

[51] P. Litvinov, O. Hasekamp, B. Cairns, and M. Mishchenko, "Reflection models for soil and vegetation surfaces from multiple-viewing angle photopolarimetric measurements," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 111, no. 4, pp. 529–539, Mar. 2010.

[52] Z. Li, Y. Zhang, H. Xu, K. Li, O. Dubovik, and P. Goloub, "The fundamental aerosol models over China region: A cluster analysis of the ground-based remote sensing measurements of total columnar atmosphere," *Geophys. Res. Lett.*, vol. 46, no. 9, pp. 4924–4932, May 2019.

[53] R. C. Levy, L. A. Remer, and O. Dubovik, "Global aerosol optical properties and application to Moderate Resolution Imaging Spectroradiometer aerosol retrieval over land," *J. Geophys. Res., Atmos.*, vol. 112, no. D13, Jul. 2007, Art. no. D13210.

[54] R. S. Fraser, R. A. Ferrare, Y. J. Kaufman, B. L. Markham, and S. Mattoo, "Algorithm for atmospheric corrections of aircraft and satellite imagery," *Int. J. Remote Sens.*, vol. 13, no. 3, pp. 541–557, Feb. 1992.

[55] T. Eck et al., "Wavelength dependence of the optical depth of biomass burning, urban, and desert dust aerosols," *J. Geophys. Res.*, vol. 104, no. 24, pp. 31333–31349, Dec. 1999.

[56] Y. J. Kaufman, A. E. Wald, L. A. Remer, B.-C. Gao, R.-R. Li, and L. Flynn, "The MODIS 2.1-*mu* channel-correlation with visible reflectance for use in remote sensing of aerosol," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 5, pp. 1286–1298, Sep. 1997.

[57] Z. Li et al., "Directional polarimetric camera (DPC): Monitoring aerosol spectral optical properties over land from satellite observation," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 218, pp. 21–37, Oct. 2018.



Bangyu Ge received the Ph.D. degree in cartography and geographical information system from the Aerospace Information Research Institute (AIR), Chinese Academy of Sciences (CAS), Beijing, China, in 2020.

From 2020 to 2022, he was a Post-Doctoral Research Associate at AIR, CAS. He is currently an Associate Professor with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources of the People's Republic of China, Beijing. His research interests include aerosol and surface retrieval from multisource satellite remote sensing data and atmospheric radiative transfer.



Xia Wang received the M.S. degree in geodesy and surveying engineering from the School of Civil Engineering, Tsinghua University, Beijing, China, in 1999.

She is currently a Professor with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources of the People's Republic of China, Beijing. She has been engaged in satellite remote sensing data processing research for a long time and has made significant achievements in theoretical method innovation, key technology breakthroughs, the development of standards and specifications for satellite data products, as well as the operational production and engineering application of satellite remote sensing data.



Zhengqiang Li received the Ph.D. degree in optics from Anhui Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, Hefei, China, in 2004.

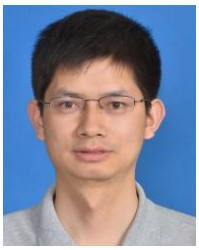
He worked at the University of Lille, Lille, France, from 2004 to 2009. He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, and the Executive Director of the State Environment Protection Key Laboratory of Satellite Remote Sensing, Beijing, China. He was involved in and completed more than 20 national key research and development programs, including China High Resolution Earth Observation Satellite Program, 973, 863, Sino-German Science Foundation, and other projects, and published more than 200 articles, four monographs, and more than 30 patents. He is mainly engaged in satellite remote sensing and atmospheric environment research.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.



Ping Zhou was born in 1980. He received the M.S. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2009 and 2016, respectively.

He is currently a Professor with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources of the People's Republic of China, Beijing, China. His research interests include photogrammetry and processing of remote sensing imagery.



Guoyuan Li received the B.S. degree in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2007, the M.S. degree in signal and information processing from the Graduate School, Chinese Academy of Sciences, Beijing, China, in 2010, and the Ph.D. degree in cartography and geographical information system from the School of Resource and Environment Sciences, Wuhan University, in 2017.

He is currently a Researcher with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources of the People's Republic of China, Beijing. His research interests include satellite laser altimetry, satellite development plan, and high-resolution stereo mapping satellites.



Weizhen Hou received the Ph.D. degree from the Graduate University of Chinese Academy of Sciences (GUCAS), Beijing, China, in 2011.

Since 2011, he has been an Assistant/Associate Professor with the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences (CAS), Beijing. From 2014 to 2016, he was a Post-Doctoral Research Associate with the University of Nebraska-Lincoln, Lincoln, NE, USA. In 2017, he joined The University of Iowa, Iowa City, IA, USA, as a Post-Doctoral Research Scholar. From 2019 to 2022, he was an Associate Professor with the Aerospace Information Research Institute, CAS. His research interests include hyperspectral and polarimetric remote sensing.

Zhenwei Qiu was born in 1985. He received the B.S. degree from the Department of Precision Machinery and Precision Instruments, University of Science and Technology of China, Hefei, China, in 2004, and the Ph.D. degree in optics from the University of Chinese Academy of Sciences, Beijing, China, in 2013.

He is currently a Research Fellow and the Director of the Optical Remote Sensing Center, Anhui Institute of Optics and Fine Mechanics, Hefei Institutes of Physical Science, Chinese Academy of Sciences (CAS), Beijing. He has led multiple national and CAS projects, including the National Natural Science Foundation, CAS key deployment projects, and CAS major instrument special projects. He is a Chief Designer or a Deputy Chief Designer of multiple high-resolution and air-based special project payload model tasks. He is leading the research on multiple model background pre-research projects. He has published more than ten journal articles and holds more than ten patents. His research focuses on aerosol particle scattering measurement, polarized optical remote sensing technologies, high-resolution satellite imaging and image enhancement techniques, calibration and correction technologies for polarized remote sensors, and engineering technologies for spaceborne sensors.

Prof. Qiu is a member of the Youth Promotion Association, CAS. He received the First Prize of Anhui Provincial Science and Technology Progress.



Chenchao Xiao received the B.S. degree from the School of Geographical Sciences, Northwest University, Xi'an, China, in 2007, and the Ph.D. degree in cartography and geographic information system from Peking University, Beijing, China, in 2017.

His research interests include remote sensing data processing and data analysis.

Shuhan Liu received the M.S. degree from the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China, in 2014.

Her research interests include remote sensing modeling and hyperspectral data preprocessing.

Qingxing Yue received the Ph.D. degree in photogrammetry and remote sensing from the School of Remote Sensing Information Engineering, Wuhan University, Wuhan, China, in 2011.

He is currently a Professor with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources of the People's Republic of China, Beijing, China. His research interests include satellite imaging simulation and satellite data processing.



Haofei Wang received the Ph.D. degree in atmospheric remote sensing from the Institute of Aerospace Information Innovation, Chinese Academy of Sciences, Beijing, China, in 2020.

He is currently an Associate Professor with the National Satellite Meteorological Center, China Meteorological Administration, Beijing. He is in charge of the calibration and validation for multiangle polarization imager onboard Fengyun Satellite. His research interests include cloud and aerosol remote sensing, as well as the simulation and calibration of optical sensors.



Yisong Xie received the Ph.D. degree in cartography and geographical information system from the Institute of Remote Sensing and Digital Earth (RADI), Chinese Academy of Sciences (CAS), Beijing, China, in 2014.

He is currently an Associate Professor with the Aerospace Information Research Institute (AIR), CAS. His research interests include retrieval of aerosol optical properties and estimation of aerosol components from polarized satellite sensors.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:04:28 UTC from IEEE Xplore. Restrictions apply.